



UI | | Agent

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PROFESSIONAL SUMMARY

Web UI Design: tao-jiayan.pages.dev Swiss Modernism
Cloudflare Pages GitHub Pages Gitee SEO JSON-LD

Computational Physics: GPA
3.76/4.0 NEGF
github.com/jiatao058-afk/my-code

Agent Application Development: AI Agent Claude
Code AI

AREAS OF EXPERTISE

- | | | | |
|---------------------------|--------------------------|--------------------------|-----------------------|
| • Python | • NumPy/SciPy/Matplotlib | • Web UI Design | • HTML/CSS/JavaScript |
| • Playwright Web Scraping | • Git Version Control | • NEGF Quantum Transport | • LaTeX |
| • Agent Applications | • Claude Code | • MATLAB | • Responsive Design |

PROFESSIONAL EXPERIENCE

Guangzhou University — Department of Physics

Graduate Research Assistant | 2025 – Present | Guangzhou, CN

- Built NEGF-based numerical framework to simulate electron transport in mesoscopic devices, modeling Hamiltonian matrices, self-energy, and transmission coefficients with Python (NumPy/SciPy/Kwant)
- Studied dephasing effects in zigzag graphene nanotubes using Buttiker virtual probe formalism, analyzing conductance dependence on device length, temperature, and dephasing strength
- Independently completed two condensed matter physics manuscripts from theoretical modeling through numerical solution to final manuscript preparation in LaTeX
- Published all computational code as open-source at github.com/jiatao058-afk/my-code for reproducibility

Independent Projects — Web & Automation

Full-Stack Developer | 2025 – Present | Remote

- Designed and built a responsive personal academic homepage (tao-jiayan.pages.dev) with SEO-optimized semantic HTML5, JSON-LD structured data, and CSS3 animations; deployed across Cloudflare Pages, GitHub Pages, and Gitee Pages
- Developed browser automation tools using Playwright framework for web data collection, automated form filling, and batch workflow processing
- Built interactive browser games (rock-paper-scissors, reaction test) using Vue 3 + TypeScript, covering full development cycle from UI design to production deployment

CERTIFICATIONS